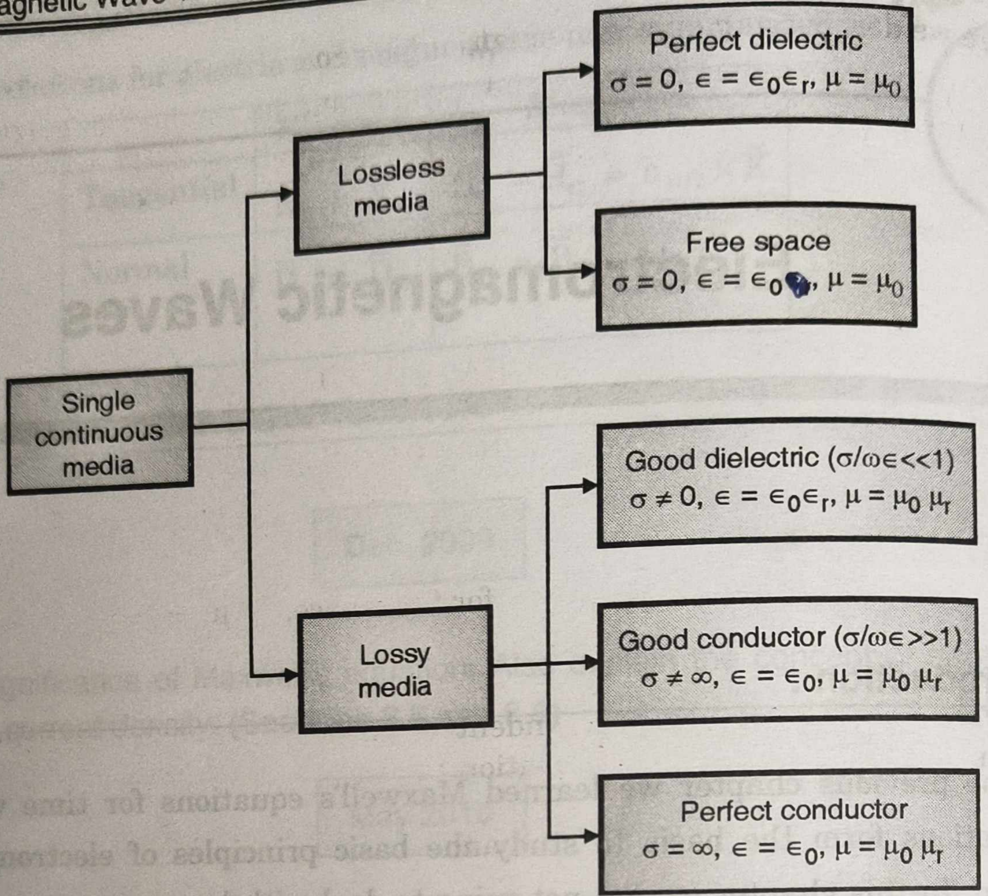

Fig. 9.1.1 : Classification of media

In this chapter we shall study how the wave propagates in a single continuous media, which can be one of the media explained above.

9.2 Wave Propagation in Lossless Media :

Lossless media is either perfect dielectric with the property,

$$\sigma = 0, \qquad \epsilon = \epsilon_0\,\epsilon_r, \qquad \mu = \mu_0\,\mu_r$$

or free space with the property

$$\sigma = 0,\ \ \epsilon = \epsilon_0,\ \mu = \mu_0$$

We will obtain the results for perfect dielectric and then putting $\mu_r = 1$ and $\epsilon_r = 1$ in the results, we get results for free space.

The term σ represents loss. For $\sigma = 0$, means no loss, so energy of the wave when propagates remain constant. We will prove it mathematically. This requires wave equations to be derived.

9.2.1 Wave Equations for Lossless Media :

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Though propagation of a wave is a physical phenomena it is explained after deriving certain mathematical expressions called **wave equations.** These are derived in this section.

Before we derive complex wave equation for general case, let us derive simple wave equation for dielectric media containing no charges and no conduction currents. To derive wave equations in dielectric we start with Maxwell's equations for dielectric, which are :

$$\nabla \times \bar{E} = -\dot{\bar{B}} \quad ...(i)$$

$$\nabla \times \bar{H} = \dot{\bar{D}} \quad ...(ii)$$

$$\nabla \cdot \bar{D} = 0 \text{ or } \nabla \cdot \bar{E} = 0 \,(\because \bar{D} = \in \bar{E}) \quad ...(iii)$$

$$\nabla \cdot \bar{B} = 0 \text{ or } \nabla \cdot \bar{H} = 0 \,(\because \bar{B} = \mu \bar{H}) \quad ...(iv)$$

Also we know that,

$$\bar{D} = \in \bar{E} \qquad \text{for free space,} \quad \in = \in_0$$

$$\bar{B} = \mu \bar{H} \qquad \text{for free space,} \quad \mu = \mu_0$$

$$\bar{J} = \sigma \bar{E} \qquad \text{for free space,} \quad \sigma = 0$$

The constants $\in$, μ and σ are independent of time. Single differentiation and double differentiation of both sides of above equations gives

$$\dot{\bar{D}} = \in \dot{\bar{E}} \quad \text{and} \quad \ddot{\bar{D}} = \in \ddot{\bar{E}} \quad ...(v)$$

$$\dot{\bar{B}} = \mu \dot{\bar{H}} \quad \text{and} \quad \ddot{\bar{B}} = \mu \ddot{\bar{H}} \quad ...(vi)$$

$$\dot{\bar{J}} = \sigma \dot{\bar{E}} \quad \text{and} \quad \ddot{\bar{J}} = \sigma \ddot{\bar{E}} \quad ...(vii)$$

We have, $\nabla \times \bar{H} = \dot{\bar{D}}$

differentiating both sides partially w. r. t time we get

$$\frac{\partial}{\partial t}(\nabla \times \bar{H}) = \frac{\partial}{\partial t}\dot{\bar{D}} = \ddot{\bar{D}}$$

(double dot is double differentiation partially w. r. t time)

i.e.

$$\nabla \times \frac{\partial}{\partial t}\bar{H} = \ddot{\bar{D}}$$

or

$$\nabla \times \dot{\bar{H}} = \ddot{\bar{D}} \quad ...(viii)$$

Similarly from,

$$\nabla \times \bar{E} = -\dot{\bar{B}}$$

We get,

$$\nabla \times \dot{\bar{E}} = -\ddot{\bar{B}} \quad ...(ix)$$

There are two types of wave equations,

(A)	Wave equation for electric field
(B)	Wave equation for magnetic field

(i) **Wave equations for electric field :** To derive wave equation for electric field we start with Equation (i).

(ii) Wave equations for magnetic field : To derive wave equation for magnetic field we start with Equation (ii)

Both electric and magnetic wave equations are in similar forms.

(A) Wave equation for electric field :

Starting with Equation (i) : $\nabla \times \bar{E} = -\dot{\bar{B}}$

Taking curl of both sides,

$$\nabla \times \nabla \times \bar{E} = -\nabla \times \dot{\bar{B}} = -\nabla \times (\mu \dot{\bar{H}}) \quad \text{...from (vi)}$$

$$= -\mu (\nabla \times \dot{\bar{H}}) = -\mu (\ddot{\bar{D}}) \quad \text{...from (viii)}$$

$$= -\mu (\epsilon \ddot{\bar{E}}) = -\mu \epsilon \ddot{\bar{E}} \quad \text{...(A)}$$

Using vector identity :

$$\nabla \times \nabla \times \bar{E} = \nabla (\nabla \cdot \bar{E}) - \nabla^2 \bar{E} \quad \text{...(x)}$$

but, $\nabla \cdot \bar{E} = 0$ ...from (iii)

then Equation (x) reduces to,

$$\nabla \times \nabla \times \bar{E} = -\nabla^2 \bar{E} \quad \text{...(B)}$$

Comparing Equation (A) and (B),

i.e. $$-\nabla^2 \bar{E} = -\mu \epsilon \ddot{\bar{E}}$$

or $$\nabla^2 \bar{E} = \mu \epsilon \ddot{\bar{E}} \quad \text{...(9.2.1)}$$

This is called as **wave equation for electric field for lossless media.**

(B) Wave equation for magnetic field :

Starting with Equation (ii) : $\nabla \times \bar{H} = \dot{\bar{D}}$

Taking curl of both sides we get,

$$\nabla \times \nabla \times \bar{H} = \nabla \times \dot{\bar{D}} = \nabla \times (\epsilon \dot{\bar{E}}) \quad \text{...from (v)}$$

$$= \epsilon (\nabla \times \dot{\bar{E}}) = \epsilon (-\ddot{\bar{B}}) \quad \text{...from (ix)}$$

$$= \epsilon (-\mu \ddot{\bar{H}}) = -\mu \epsilon \ddot{\bar{H}} \quad \text{...(C)}$$

Using vector identity,

$$\nabla \times \nabla \times \bar{H} = \nabla (\nabla \cdot \bar{H}) - \nabla^2 \bar{H} \quad \text{...(xi)}$$

But $\nabla \cdot \bar{H} = 0$ ...from (iv)

Then Equation (xi) reduces to,

$$\nabla \times \nabla \times \bar{H} = -\nabla^2 \bar{H} \quad \text{...(D)}$$

Comparing Equations (C) and (D) we get

$$-\nabla^2 \bar{H} = -\mu \in \ddot{\bar{H}} \qquad \nabla^2 \bar{H} = \mu \in \ddot{\bar{H}} \quad ...(9.2.2)$$

"Equation (9.2.2) is expression for wave equation for magnetic field, for lossless media."

Noticing the difference in properties of dielectric and free space, just by replacing μ and $\in$ by μ_0 and $\in_0$, wave equations for free space can be obtained.

Wave equations for lossless media	
Dielectric	**Free space**
$\nabla^2 \bar{E} = \mu\in \ddot{\bar{E}}$	$\nabla^2 \bar{E} = \mu_0\in_0\ddot{\bar{E}}$
$\nabla^2 \bar{H} = \mu\in \ddot{\bar{H}}$	$\nabla^2 \bar{H} = \mu_0\in_0\ddot{\bar{H}}$

Using these wave equations we can study propagation field, for lossless media.

9.2.2 Properties of Plane Waves :

The waves which have studied in previous articles all are uniform plane waves. In order to understand what are these waves and the importance of it, let us see its properties first.

Wavefront :

A wavefront is defined as the locus of points having the same phase.

We know the term phase constant (β) expressed in rad/m. When the wave travels in a space the phase angle of the wave

$$\text{Phase angle or phase} = \beta \times \text{distance}$$

is charging continuously since the distance is changing continuously. The waves travelled same distance from the source (for example antenna) will undergo same change in phase. If we consider a surface passing through all these equiphase points, the surface is called as **wavefront.**

In case of isotropic antenna, waves are radiated equally in all directions. If we consider this antenna as a centre of a sphere of radius R, then every point on the sphere will be at a same distance from an antenna. Since distance travelled by all waves to reach the sphere is same, all waves will undergo same change in phase angle (equal to βR). These waves are said to have same phase and the sphere is called as **spherical wavefront.** Or the waves are called as spherical waves.

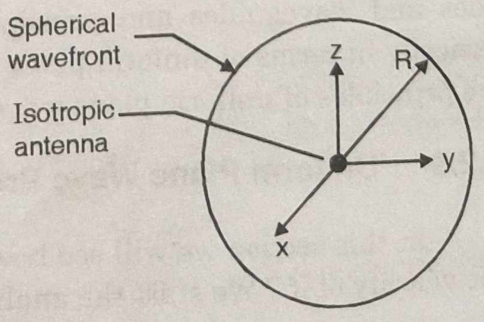

Fig. 9.2.1 : Showing spherical wavefront

In order to understand the uniform plane waves now you should know plane waves. It is defined as :

A plane wave is a constant frequency waves whose wavefornts are infinite parallel planes of constant amplitude normal to the phase velocity vector (direction of propagation).

By extension, the term is also used to describe waves that are approximately plane waves in a localized region of space. Plane waves does not exist in practice because a source of infinite in extent would be required to creat it and practical wave sources are always finite in extent. But if we are far enough from a source, the wavefront becomes almost spherical and a very small portion of the surface of a giant sphere is very nearly a plane. Plane wave is also defined as :

If electric and magnetic field of a wave lies in a plane perpendicular to the direction of wave travel, the wave is called as plane wave.

Now let us see what is uniform plane wave.

Uniform plane waves :

If the field has the same direction and magnitude at every point in any plane perpendicular to the direction of wave travel, the wave is called as uniform plane wave. This is as shown in Fig. 9.2.2.

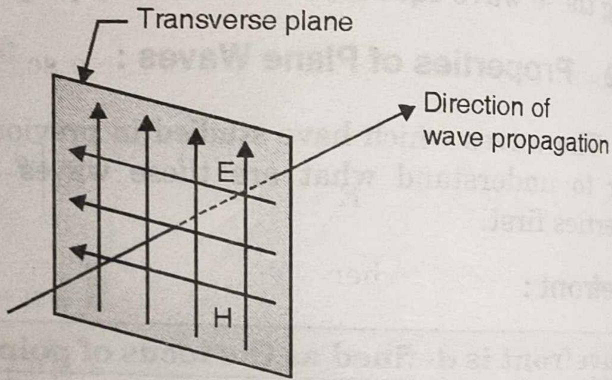

Fig. 9.2.2

Why uniform plane waves ?

A simple and very useful type of wave that serves a building block in the study of electromagnetic waves consists of electric and magnetic fields that are perpendicular to each other and to the direction of propagation of wave. Also they are uniform in amplitudes perpendicular to the direction of propagation. These waves are nothing but uniform plane waves.

Further more the principle of guiding of electromagnetic waves along transmission lines and waveguides and principles of many other wave phenomena can be studied basically in terms of uniform plane waves. Hence it is very important that we understand the principles of uniform plane wave propagation.

9.2.3 Uniform Plane Wave Propagation (Solution of Wave Equation) :

In this section we will see how the wave propagates in perfect dielectric and what is the velocity of it ? We start the analysis by wave equations derived in the last section.

The wave equations for electric and magnetic fields are :

$$\nabla^2 \bar{E} = \mu \in \ddot{\bar{E}} \quad \text{...(i)}$$

and

$$\nabla^2 \bar{H} = \mu \in \ddot{\bar{H}} \quad \text{...(ii)}$$

As the form of Equations (i) and (ii) is same, once we make the analysis of one equation, the result can be applied to other equation directly.

Let the electric and magnetic field intensities are defined in cartesian system as

$$\bar{E} = E_x \bar{a}_x + E_y \bar{a}_y + E_z \bar{a}_z$$

$$\bar{H} = H_x \bar{a}_x + H_y \bar{a}_y + H_z \bar{a}_z$$

Starting with Equation (i)

$$\nabla^2 \bar{E} = \mu \in \ddot{\bar{E}}$$

This wave equation gives rise to three **scalar wave equation**. Putting value of $\bar{E}$ in this equation gives :

$$\nabla^2 E_x \bar{a}_x + \nabla^2 E_y \bar{a}_y + \nabla^2 E_z \bar{a}_z = \mu\in \left[\frac{\partial^2 E_x}{\partial t^2}\bar{a}_x + \frac{\partial^2 E_y}{\partial t^2}\bar{a}_y + \frac{\partial^2 E_z}{\partial t^2}\bar{a}_z\right]$$

Two vectors are equal when x, y and z components of vectors are identical, comparing two sides, we get scalar wave equations as :

$$\nabla^2 E_x = \mu\in \frac{\partial^2 E_x}{\partial t^2},$$

$$\nabla^2 E_y = \mu\in \frac{\partial^2 E_y}{\partial t^2},$$

and

$$\nabla^2 E_z = \mu\in \frac{\partial^2 E_z}{\partial t^2}$$

The solution of these three differential equations can be obtain on similar lines. Let's consider for the sake of simplicity that the time varying electric field intensity is a function of x only, so that all ∂/∂y and ∂/∂z concern with E are zero. The value of ∇^2 is also simplified as shown below :

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \Rightarrow \frac{\partial^2}{\partial x^2}$$

The above three equations containing $\nabla^2 E_x$, $\nabla^2 E_y$ and $\nabla^2 E_z$ can be written using reduced expression for ∇^2 as

$$\frac{\partial^2 E_x}{\partial x^2} = \mu\in \frac{\partial^2 E_x}{\partial t^2} \quad \text{...(iii)}$$

$$\frac{\partial^2 E_y}{\partial x^2} = \mu\in \frac{\partial^2 E_y}{\partial t^2} \quad \text{...(iv)}$$

$$\frac{\partial^2 E_z}{\partial x^2} = \mu\epsilon \frac{\partial^2 E_z}{\partial t^2} \quad ...(v)$$

Notice that Equations (iii), (iv) and (v) are scalar wave equations. According to Maxwell's equation for lossless media, we have

$$\nabla \cdot \bar{D} = 0 \quad \text{then} \quad \nabla \cdot \bar{E} = 0 \quad (\therefore \bar{D} = \epsilon \bar{E})$$

Expanding div.E using expression for E in cartesian system, we get

$$\nabla \cdot \bar{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

$$\therefore \quad \nabla \cdot \bar{E} = \frac{\partial E_x}{\partial x} \Rightarrow 0 \quad (\because \partial/\partial y = \partial/\partial z = 0)$$

In the above equation differentiation of E_x is zero means E_x is either constant or zero. But if we accept that E_x is constant then we are not sticking with the condition that E is a variable of x. Thus, we are forced to conclude that

$$E_x = 0 \quad ...(9.2.3)$$

The above equation shows that the **electric field intensity varying w.r.t x does not have the x component**. So that the electric field vector which we initially assumed in terms of E_x, E_y and E_z becomes,

$$\bar{E} = E_y \bar{a}_y + E_z \bar{a}_z \quad ...(vi)$$

Also Equation (iii) is totally vanished. Now consider Equations (iv) and (v), which are the differential Equations for E_y and E_z having same form. By solving Equation (iv) we get electric field E_y. The form of Equations (iv) and (v) is same therefore once we obtain solution for E_y, the solution for E_z can very easily be obtained.

The **general solution** of Equation (iv) can be specified as

$$\boxed{E_y = f(x - v_0 t)} \quad ...(vii)$$

This function may be any function of $(x - v_0 t)$ e.g. it may be $\sqrt{x - v_0 t}$ or $e^{(x - v_0 t)}$ or $\sin(x - v_0 t)$. We can easily check that for any of this function, E_y is satisfying Equation (iv) provided that

$$v_0 = \frac{1}{\sqrt{\mu\epsilon}} \quad ...(9.2.4)$$

For the sake of simplicity assume

$$\boxed{E_y = \sin(x - v_0 t)} \quad ...(9.2.5)$$

If we plot this function w. r. t. x at different timings, we get

at $t = 0$, $\quad E_y = \sin x$

at $t = t_1$, $t_1 > 0$, $\quad E_y = \sin(x - x_1)$ where $x_1 = v_0 t_1$

at $t = t_2$, $t_2 > t_1 > 0$, $\quad E_y = \sin(x - x_2)$ where $x_2 = v_0 t_2$ etc.

All these sinusoidal functions are plotted in Fig. 9.2.3.

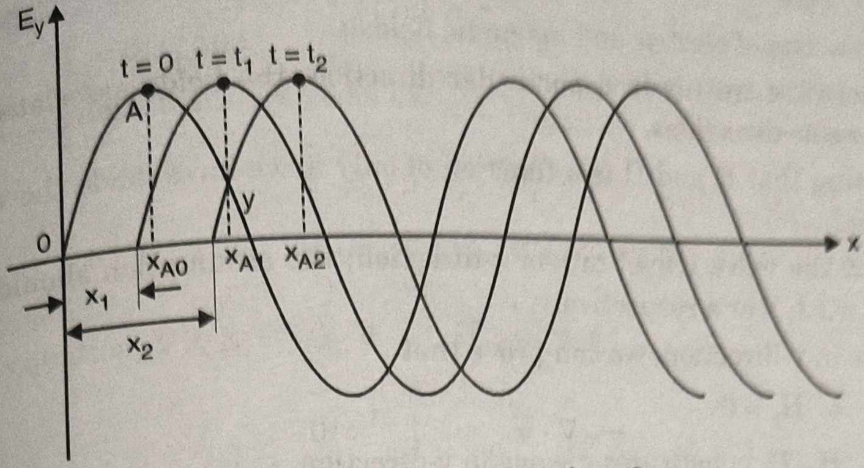

Fig. 9.2.3 : Showing propagation of wave

By observing variation of E_y w.r.t x at different timings we can conclude that at time $t = 0$, the sinusoidal positive starts at $x = 0$. At $t = t_1$ the sinusoidal positive going starts at $x = x_1 = v_0\, t_1$. Similarly at time t_2, sinusoid starts at $x = x_2 = v_0\, t_2$.

If we consider any point on the waveform at $t = 0$, e.g. A, it is at x_{A0} position. Then at $t = t_1$, it is at x_{A1} ; at $t = t_2$, its position changes to x_{A2}. Here we observe that as time progresses, point A moves forward in x direction. This is true for all points on the first waveform or in general the phenomenon of variation of E_y is propagating in x direction.

We have $x = v_0 t$, where x is measured in meters while t is measured in seconds, then the unit of v_0 is m/s. Thus v_0 represents a velocity of wave.

The similar type of analysis can be made for Equation (v) to show that the variation of E_z w.r.t. x also constitute a wave.

The above analysis of Equation (i) concludes that for E as a function of x only,

- $E_x = 0$
- Variation of E_y constitutes a wave travelling in x-direction.
- Variation of E_z constitutes a wave travelling in x-direction.

On the similar lines if we make analysis of Equation (ii), then we can write for H varying as a function of x only,

- $H_x = 0$
- Variation of H_y constitutes a wave travelling in x-direction.
- Variation of H_z constitutes a wave travelling in x-direction.

Note :

- A wave consists of electric and magnetic fields.
- When the wave travels in a particular direction, the fields associated with it also travel in same directions.
- By assuming that E and H is a function of only x, we have made the wave travel in x-direction.
- If we want the wave travel in y-or z-direction; the assumption should be, E and H vary only w.r.t, y or z respectively.
- For a wave in y-direction, we can prove that

$$E_y = H_y = 0$$

and, E_x, E_z, H_x, H_z constitutes a wave in y-direction.

- Similarly, for a wave in z-direction.

$$E_z = H_z = 0$$

and, E_x, E_y, H_x, H_y constitutes a wave in z-direction.

- All field components travel with a velocity,

$$\upsilon_0 = \frac{1}{\sqrt{\mu\epsilon}}$$

9.2.4 Intrinsic Impendence or Characteristic Impedance of Lossless Medium :

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We have the unit of electric and magnetic field as(V/m) and (A/m) respectively. If we take the ratio.

$$\frac{E}{H} \Rightarrow \frac{V/m}{A/m} = V/A, \quad \text{which is a unit of impedance .}$$

At some point on the circuit board, having the knowledge of voltage and current at that point, the impedance at that point can be determined simply by taking the ratio of them, this idea can be extended further.

The wave in a media is associated with the electric and magnetic field with it. At some point in the media, by taking the ratio of E and H, the impedance of media at that point can be obtained. This impedance is called as **intrinsic impedance** denoted by (η). It is defined as,

Intrinsic impedance :

It is the ratio of the magnitudes of E and H for a plane (TEM) wave in an unbounded medium. Mathematically,

$$\boxed{\eta = \text{Intrinsic impedance} = \frac{\text{Magnitude of E}}{\text{Magnitude of H}}}$$

For a wave propagating in x- direction we have

$$E_x = H_x = 0 \quad \text{and} \quad \partial/\partial y = \partial/\partial z = 0$$

Then electric and magnetic fields are given by,

$$\bar{E} = E_y \bar{a}_y + E_z \bar{a}_z \quad \text{...(i)}$$

$$\bar{H} = H_y \bar{a}_y + H_z \bar{a}_z \quad \text{...(ii)}$$

Using Maxwell's equation, $\nabla \times \bar{E} = -\dot{\bar{B}} = -\mu \dot{\bar{H}}$, we get

$$\nabla \times E = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \partial/\partial x & 0 & 0 \\ 0 & E_y & E_z \end{vmatrix} = -\mu \frac{\partial}{\partial t} [H_y \bar{a}_y + H_z \bar{a}_z]$$

Solving determinant in above equation we get,

$$-\frac{\partial E_z}{\partial x} \bar{a}_y + \frac{\partial E_y}{\partial x} \bar{a}_z = -\mu \left[\frac{\partial H_y}{\partial t} \bar{a}_y + \frac{\partial H_z}{\partial t} \bar{a}_z\right] \quad \text{...(a)}$$

Similarly, using Maxwell's equation for $\nabla \times \bar{H} = \dot{\bar{D}} = \in \dot{\bar{E}}$ we get,

$$-\frac{\partial H_z}{\partial x} \bar{a}_y + \frac{\partial H_y}{\partial x} \bar{a}_z = \in \left[\frac{\partial E_y}{\partial t} \bar{a}_y + \frac{\partial E_z}{\partial t} \bar{a}_z\right] \quad \text{...(b)}$$

From Equations, (a) and (b) we have following relation between components of E and components of H

$$\frac{\partial E_z}{\partial x} = \mu \frac{\partial H_y}{\partial t} \quad \text{...(iii)}$$

$$\frac{\partial E_y}{\partial x} = -\mu \frac{\partial H_z}{\partial t} \quad \text{...(iv)}$$

$$-\frac{\partial H_z}{\partial x} = \in \frac{\partial E_y}{\partial t} \quad \text{...(v)}$$

$$\frac{\partial H_y}{\partial x} = \in \frac{\partial E_z}{\partial t} \quad \text{...(vi)}$$

Now if we assume solution for E_y as, $E_y = f(x - v_0 t)$. Then differentiation w.r.t. x gives

$$\frac{\partial E_y}{\partial x} = \frac{\partial f(x - v_0 t)}{\partial (x - v_0 t)} \cdot \frac{\partial (x - v_0 t)}{\partial x} = f'$$

where

$$f' = \frac{\partial f(x - v_0 t)}{\partial (x - v_0 t)}$$

Using this in Equation (iv),

$$f' = -\mu \frac{\partial H_z}{\partial t} \quad \text{...(vii)}$$

Integrating w.r.t t, we get,

$$H_z = -\frac{1}{\mu} \int f' dt$$

Differentiating E_y w.r.t. t gives,

$$\frac{\partial E_y}{\partial t} = \frac{\partial f(x - v_0 t)}{\partial (x - v_0 t)} \cdot \frac{\partial (x - v_0 t)}{\partial t} = (-\upsilon_0) f'$$

or

$$E_y = \int -v_0 f' \, dt = -\upsilon_0 \int f' \, dt \quad \text{...(viii)}$$

Dividing Equation (viii) by Equation (vii) we get,

$$\frac{E_y}{H_z} = \frac{v_0}{1/\mu} = \mu v_0 \qquad \text{But,} \quad v_0 = \frac{1}{\sqrt{\mu \epsilon}}$$

$$\therefore \quad \frac{E_y}{H_z} = \mu \cdot \frac{1}{\sqrt{\mu \epsilon}} = \sqrt{\frac{\mu}{\epsilon}} = \eta \quad \text{...(ix)}$$

where

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

By assuming expression for E_z as $\quad E_z = f(x - v_0 t)$

and giving the similar treatment as for E_y we get,

$$\frac{E_z}{H_y} = -\sqrt{\frac{\mu}{\epsilon}} = -\eta \quad \text{...(x)}$$

Combining Equations (ix) and (x) we get,

$$\frac{E_y}{H_z} = -\frac{E_z}{H_y} = \eta$$

or

$$E_y = \eta H_z$$

and

$$E_z = -\eta H_y$$

Using Equation (i)

$$\bar{E} = E_y \bar{a}_y + E_z \bar{a}_z = \eta H_z \bar{a}_y - \eta H_y \bar{a}_z$$

Taking mod of both sides :

i.e.

$$|\bar{E}| = \eta \sqrt{H_y^2 + H_z^2}$$

i.e.

$$|\bar{E}| = \eta |\bar{H}|$$

or

$$\boxed{\eta = \frac{|\bar{E}|}{|\bar{H}|} = \sqrt{\frac{\mu}{\epsilon}}} \quad \text{...(9.2.6)}$$

As E is measured in Volts/m and H is measured in Amp/m, then the unit of η will be volts / Amp i.e. Ω and it is called as **intrinsic impedance of the media.**

By putting $\mu = \mu_0 = 4\pi \times 10^{-7}$ and $\epsilon = \epsilon_0 = 10^{-9}/36\pi$ we get intrinsic impedance of free space

$$\boxed{\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{(10^{-9}/36\pi)}} \simeq 120\,\pi = 377(\Omega)}$$

The term $\bar{a}_\beta$ is also called as phase vector or propagation vector. To understand how the use this formula consider a simple example.

Let the wave is in x direction and the electric field has E_y component, then

$$\bar{a}_E = \bar{a}_y \text{ and } \bar{a}_\beta = \bar{a}_x$$

Now suppose we have to find magnetic field. To obtain the direction of magnetic field, we use the formula.

$$\bar{a}_E \times \bar{a}_H = \bar{a}_\beta$$

$$\bar{a}_y \times \bar{a}_H = \bar{a}_x$$

Knowing the cross product of unit vectors,

$$\bar{a}_y \times \bar{a}_z = \bar{a}_x$$

we get the direction of $\bar{a}_H$.

This gives $\bar{a}_H = \bar{a}_z$. To find the magnitude of magnetic field we use,

$$\frac{E}{H} = \eta$$

or

$$H = \frac{E}{\eta}$$

Knowing magnitude and direction we can write H.

$$\bar{H} = \frac{E}{\eta}\bar{a}_z$$

If magnetic field $\bar{H}$ is given and we are interested in electric field $\bar{E}$, we use same procedure.

9.2.6 Helmholtz Equations using Maxwell's Equations :

A) Helmholtz equation for electric field

B) Helmholtz equation for magnetic field

We have Maxwell's equations for harmonically varying fields in lossless media ($\bar{J} = 0$ and $\rho_v = 0$) as follows :

$\nabla \times \bar{E} = -\dot{\bar{B}} = -\frac{dB}{dt}$

$= -j\omega\bar{B}$

$$\nabla \times \bar{E} = -j\omega\bar{B} = -j\omega\mu\bar{H} \quad (\because \bar{B} = \mu\bar{H}) \qquad \text{...(i)}$$

$$\nabla \times \bar{H} = j\omega\bar{D} = j\omega\epsilon\bar{E} \quad (\because \bar{D} = \epsilon\bar{E}) \qquad \text{...(ii)}$$

$$\nabla \cdot \bar{D} = 0 \qquad \text{i.e. } \nabla \cdot \bar{E} = 0 \qquad \text{...(iii)}$$

$$\nabla \cdot \bar{B} = 0 \qquad \text{i.e. } \nabla \cdot \bar{H} = 0 \qquad \text{...(iv)}$$

A) Helmholtz equation for electric field :

Taking curl of Equation (i), $\nabla \times \nabla \times \bar{E} = \nabla \times (-j\omega\mu\bar{H}) = -j\omega\mu\,(\nabla \times \bar{H})$

Putting value of $\nabla \times \bar{H}$ from Equation (ii),

$$\nabla \times \nabla \times \bar{E} = -j\omega\mu\,(j\omega\epsilon\bar{E}) = -j^2\omega^2\mu\epsilon\bar{E}$$

i.e.

$$\nabla \times \nabla \times \bar{E} = \omega^2\mu\epsilon\bar{E} \qquad (\because j^2 = -1) \qquad \text{...(v)}$$

From vector identity we have,

$$\nabla \times \nabla \times \bar{E} = \nabla(\nabla \cdot \bar{E}) - \nabla^2\bar{E}$$

But from Equation (iii), $\nabla \cdot \bar{E} = 0$ then above equation reduces to,

$$\nabla \times \nabla \times \bar{E} = -\nabla^2\bar{E} \qquad \text{...(vi)}$$

Comparing Equations (v) and (vi),

$$\nabla \times \nabla \times \bar{E} = \omega^2\mu\epsilon\bar{E} = -\nabla^2\bar{E}$$

i.e.

$$\boxed{\nabla^2\bar{E} = -\omega^2\mu\epsilon\bar{E}} \qquad \text{...(9.2.7)}$$

This is the **Helmholtz equation for electric field.**

B) Helmholtz equation for magnetic field :

Taking curl of Equation (ii), $\nabla \times \nabla \times \bar{H} = \nabla \times (j\omega\epsilon\bar{E})$

$$= j\omega\epsilon\,(\nabla \times \bar{E})$$

Putting value of $\nabla \times \bar{E}$ from Equation (i),

$$\nabla \times \nabla \times \bar{H} = j\omega\epsilon\,(-j\omega\mu\bar{H}) = -j^2\omega^2\mu\epsilon\bar{H}$$

i.e.

$$\nabla \times \nabla \times \bar{H} = \omega^2\mu\epsilon\bar{H} \qquad \text{...(vii)}$$

Comparing Equation (vii) with Equation (viii),

$$\nabla \times \nabla \times \bar{H} = \omega^2\mu\epsilon\bar{H} = -\nabla^2\bar{H}$$

i.e.

$$\boxed{\nabla^2\bar{H} = -\omega^2\mu\epsilon\bar{H}} \qquad \text{...(9.2.8)}$$

This is the **Helmholtz equation for magnetic field.**

Significance of Helmholtz equations is discussed in the next section.

9.2.7 Solution of Helmholtz Equations for Lossless Media :

In the section 9.2.3 we learned uniform plane wave propagation in losseless media, for which free space is a special case. In this medium the conduction current is almost absent in comparison with the displacement current. Such a medium may be treated as a perfect dielectric or lossless medium ($\sigma = 0$).

In all previous sections we made analysis in space domain and in time domain. Instead of working in time domain if we work in frequency domain the analysis is more simplified. For this purpose we start the analysis by using Helmholtz equations.

For the uniform plane wave travelling in x-direction, the electric and magnetic field intensities are varying w.r.t x only. We have Helmholtz equations as,

$$\nabla^2 \bar{E} = -\omega^2 \mu\epsilon \bar{E} = -\beta^2 \bar{E} \quad \text{...(i)}$$

$$\nabla^2 \bar{H} = -\omega^2 \mu\epsilon \bar{H} = -\beta^2 \bar{H} \quad \text{...(ii)}$$

where, $\beta = \omega\sqrt{\mu\epsilon}$

For variation in only x direction the Helmholtz equations can be changed by using

$$E_x = H_x = 0 \text{ and all } \frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0$$

Thus, for the electric field intensity we get,

$$\frac{\partial^2 E_y}{\partial x^2} = -\beta^2 E_y \quad \text{...(iii)}$$

and

$$\frac{\partial^2 E_z}{\partial x^2} = -\beta^2 E_z \quad \text{...(iv)}$$

Considering variation of y component only

$$\frac{\partial^2 E_y}{\partial x^2} = -\beta^2 E_y \quad \text{...(v)}$$

Equation (v) is a second order differential equation whose solution can be assumed having the form

$$E_y = E_{y0}\, e^{-j\beta x} \quad \text{...(vi)}$$

This equation gives us how E_y is varying w.r.t. x. The corresponding sinusoidal time varying field is obtained by multiplying $E_y(x)$ by $e^{j\omega t}$ and then taking its real part or imaginary part. The time variation is obtained as,

$$E_y(x,t) = E_{y0}\, e^{-j\beta x} \times e^{j\omega t} = E_{y0} \times e^{j(\omega t - \beta x)} \quad \text{...(vii)}$$

The sinusoidal variation if obtained by taking the real part of Equation (vii)

$$\tilde{E}_y(x, t) = \text{R.P.}\{E_y(x, t)\} = E_{y0} \cos(\omega t - \beta x) \quad \text{...(viii)}$$

The plot of $\tilde{E}_y(x,t)$ for different timings is as shown in Fig. 9.2.4.

A) Velocity of a wave
B) Phase constant and wavelength
C) Intrinsic impedance or wave impedance

A) Velocity of a wave :

Fig. 9.2.4 shows that the wave is travelling in + x direction. If we fix our attention on a particular point (a point of a particular phase) on the wave, we set

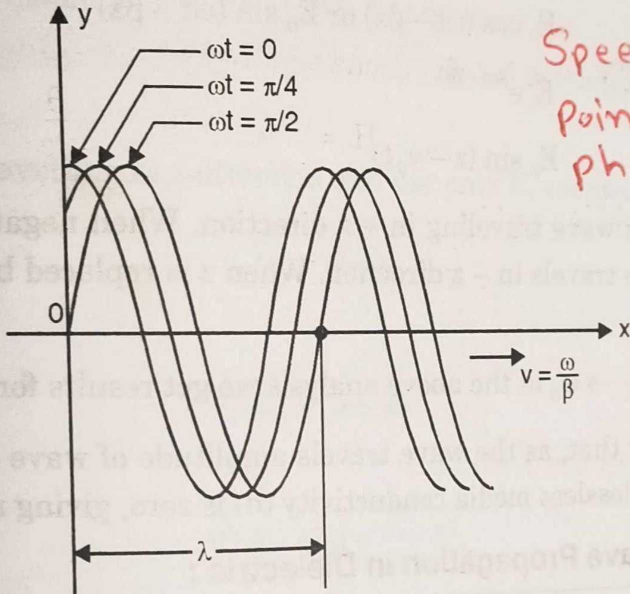
$$\cos(\omega t - \beta x) = \text{constant} \quad \text{or} \quad \omega t - \beta x = \text{constant}$$

Then the differentiation of it w.r.t time gives,

$$\omega - \beta \frac{dx}{dt} = 0 \quad \text{or} \quad \frac{dx}{dt} = \frac{\omega}{\beta} \qquad ...(ix)$$

Since x represents a distance, then its differentiation with time represents a velocity. Thus,

$$v = \frac{\omega}{\beta} \qquad ...(9.2.9)$$



Speed at which a point of constant phase propagates.

Fig. 9.2.4 : Showing propagation of plane wave in x-direction

This velocity of some point in the sinusoidal waveform is called the **phase velocity**. Putting value of β as $\omega\sqrt{\mu\epsilon}$ we get,

$$v = \frac{1}{\sqrt{\mu\epsilon}} \qquad ...(9.2.10)$$

B) Phase constant and wavelength :

β = Rate at which the phase of a wave changes along direction of propagation

The constant β is called the phase shift constant and is a measure of the phase shift in radians per unit length. Another important quantity observed with wave is wavelength λ, defined as the distance over which the sinusoidal waveform passes through a full cycle of 2π radians. Thus, we can write

$$\beta\lambda = 2\pi \qquad ...(9.2.11)$$

or

$$\lambda = 2\pi / \beta$$

Using the value of β from Equation (9.2.9) and $\omega = 2\pi f$, we get

$$\lambda = \frac{2\pi}{\omega/v} = \frac{2\pi v}{2\pi f}$$

or

$$\lambda = v/f \qquad ...(9.2.12)$$

In general we can conclude that for the sinusoidal time variation, variation w.r.t. space is also sinusoidal.

C) Intrinsic impedance or wave impedance :

The expression for intrinsic impedance remain same as ;

$$\eta = \sqrt{\mu/\epsilon}$$

Before we start solving some more examples on wave propagation in free space, remember the following forms of a uniform plane wave :

- $E_0 \cos(\omega t - \beta z)$ or $E_0 \sin(\omega t - \beta z)$
- $E_0 e^{j(\omega t - \beta z)}$ ...(9.2.13)
- $E_0 \sin(z - v_0 t)$

All these forms represent wave traveling in + z direction. When negative sign is changed by positive sign, the wave travels in – z direction. When z is replaced by x, wave travels in x direction and so on.

By replacing $\mu \rightarrow \mu_0$ and $\epsilon \rightarrow \epsilon_0$ in the above analysis we get results for free space.

From Fig. 9.2.4 it is clear that, as the wave travels amplitude of wave remain constant. It is due to the fact that for lossless media conductivity (σ) is zero, giving no loss of energy.

9.2.8 Summary of Wave Propagation in Dielectric :

I.	Wave equations	(i) For electric : $\nabla^2 \bar{E} = \mu\epsilon \ddot{\bar{E}}$ (ii) For magnetic : $\nabla^2 \bar{H} = \mu\epsilon \ddot{\bar{H}}$
II.	TEM wave in general	(i) $\bar{E} \cdot \bar{H} = 0$ (ii) $\bar{E} \times \bar{H} \Rightarrow$ in the direction of propagation.
III.	Field components for a wave in x-direction	(i) $E_x = H_x = 0$ (ii) E_y, E_z, H_y, H_z represents a wave in x-direction.
IV.	Relation between field components.	$\frac{E_y}{H_z} = -\frac{E_z}{H_y} = \eta$

V.	Velocity of a wave	$v = \frac{1}{\sqrt{\mu\epsilon}}$
VI.	Intrinsic impendence	$\eta = \sqrt{\mu / \epsilon}$
VII.	Helmholtz equations	(i) For electric : $\nabla^2 \bar{E} = -\omega^2 \mu\epsilon \bar{E}$ (ii) For magnetic : $\nabla^2 \bar{H} = -\omega^2 \mu\epsilon \bar{H}$
VIII.	Properties of wave	(i) Attenuation, $\alpha = 0$ (ii) Phase constant, $\beta = \omega\sqrt{\mu\epsilon}$ (iii) Phase velocity, $v = \omega / \beta = \frac{1}{\sqrt{\mu\epsilon}}$ (iv) Wave length, $\lambda = 2\pi / \beta = v/f$

Example 9.2.2 : Give solution to the wave equation in perfect dielectric for a wave travelling in z-direction, which has only x-component of E-field.

MU - May 2010, 10 Marks

Solution :

Given : Wave is travelling in z-direction and has only E_x component of field.

$$\bar{E} = E_x \bar{a}_x$$

The wave equation

$$\nabla^2 \bar{E} = \mu \in \ddot{\bar{E}}$$

reduces to

$$\nabla^2 E_x = \mu \in \frac{\partial^2}{\partial t^2} E_x \quad \text{...(A)}$$

Also for a wave in z-direction, $\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0.$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \rightarrow \frac{\partial^2}{\partial z^2}$$

Now Equation (A) becomes

$$\frac{\partial^2 E_x}{\partial z^2} = \mu\epsilon \frac{\partial^2 E_x}{\partial t^2}$$

The solution of this equation will be

$$E_x = \sin(z - V_0 t) \quad \text{...(B)}$$

where,

$$V_0 = \frac{1}{\sqrt{\mu\epsilon}} = \text{velocity of a wave}$$

Equation (B) is the required solution.

Example 9.2.3 : A uniform plane waves propagates in a medium with $\epsilon_r = 4, \sigma = 0, \mu_r = 1$. The E-field has only x-component which is sinusoidal with a frequency of 100 MHz and has maximum value of 0.1 mV/m at t = 0, z = 1/8. Find instantaneous expression for E(z, t) and H(z, t). Also find the location where E_x is positive maximum at t = 10 ns. **MU - May 2010, 10 Marks**

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Solution :

Given : $\epsilon_r = 4, \quad \sigma = 0, \quad \mu_r = 1$

$$f = 100 \text{ MHz} \quad \rightarrow \omega = 2\pi f = 200\,\pi \times 10^6 \text{ rad / m}$$

$$\beta = \omega\sqrt{\mu\epsilon} = 200\,\pi \times 10^6 \sqrt{\mu_0 \epsilon_0 \epsilon_r} = \frac{200\,\pi \times 10^6}{3 \times 10^8}\sqrt{4}$$

$$\rightarrow \beta = \frac{4\pi}{3} \text{ (rad/m)}$$

If we consider expression for E as

$$E_x(t) = E_{max} \cos(\omega t - \beta z) = 0.1 \times 10^{-3} \cos(\omega t - \beta z) \quad \text{...(i)}$$

$$\text{At } t = 0 \rightarrow E_x(0) = 0.1 \times 10^{-3} \cos(-\beta z) = 0.1 \times 10^{-3} \cos(\beta z)$$

$$= 0.1 \times 10^{-3} \cos\left(\frac{4\pi}{3} z\right)$$

It is given that E has maximum value of 0.1 mV / m at z = 1/8. That is $E_x(0)$ at z = 1/8 must be 0.1×10^{-3}. Putting the value of z,

$$E_x(0) = 0.1 \times 10^{-3} \cos\left(\frac{4\pi}{3} \times \frac{1}{8}\right) = 0.1 \times 10^{-3} \cos\left(\frac{\pi}{6}\right) \neq 0.1 \times 10^{-3}$$

Thus Equation (i) requires modification. Let

$$E_x(t) = 0.1 \times 10^{-3} \cos(\omega t - \beta z + \theta)$$

We find θ by using the fact that for $E_x(t)$ to be maximum cosine function should have 1 value. For this the bracketed term must be zero or 2π etc.

$$\cos(\omega t - \beta z + \theta) = 1 \rightarrow \omega t - \beta z + \theta = 0$$

Putting the values of t = 0, β and z,

$$0 - \frac{4\pi}{3} \times \frac{1}{8} + \theta = 0$$

$$\rightarrow \theta = \frac{\pi}{6}$$

So the expression for E is

$$E(z, t) = 0.1 \times 10^{-3} \cos\left(200\,\pi \times 10^6\, t - \frac{4\pi}{3} z + \frac{\pi}{6}\right) = E_x(z, t)$$

For a wave in z-direction E and H are related using

$$\frac{E_x}{H_y} = \eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} = \frac{120\,\pi}{\sqrt{\epsilon_r}} = 60\,\pi\ (\Omega)$$

$$H_y(z, t) = \frac{E_x}{60\pi} = \frac{0.1 \times 10^{-3}}{60\pi} \cos\left(200\,\pi \times 10^6\, t - \frac{4\pi}{3} z + \theta\right)$$

or

To find the position where E_x is positive maximum at $t = 10$ ns, set the cosine term equal to 1.

$$\cos\left(200\,\pi \times 10^6 \times 10 \times 10^{-9} - \frac{4\pi}{3} z + \frac{\pi}{6}\right) = 1$$

$$\rightarrow 200\,\pi \times 10^6 \times 10 \times 10^{-9} - \frac{4\pi}{3} z + \frac{\pi}{6} = 0$$

$$\rightarrow 2\pi - \frac{4\pi}{3} z + \frac{\pi}{6} = 0$$

$$\rightarrow \frac{4\pi}{3} z = 2\,\pi + \frac{\pi}{6} = \frac{13\,\pi}{6}$$

$$\rightarrow z = \frac{13}{8}\ \text{(m)}$$

9.3 Wave Propagation in Free Space :

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MU - Dec. 2005, Dec. 2006, May 2008

Look at the properties of dielectric media and free space.

For dielectric : $\sigma = 0, \quad \epsilon = \epsilon_0 \epsilon_r, \quad \mu = \mu_0 \mu_r$

For free space : $\sigma = 0, \quad \epsilon = \epsilon_0, \quad \mu = \mu_0$

By replacing ϵ by ϵ_0 and μ by μ_0 we obtain the results for free space from wave propagation in dielectric media (Refer section (9.2.8)).

(1) **Wave equations are :**

$$\nabla^2 \bar{E} = \mu_0 \epsilon_0 \ddot{\bar{E}} \quad \text{and} \quad \nabla^2 \bar{H} = \mu_0 \epsilon_0 \ddot{\bar{H}}$$

(2) **For a wave in x-direction :**

$$E_x = H_x = 0$$

and, E_y, E_z, H_y, H_z constitutes a traveling wave in + x direction.

(3) **Velocity of a wave :**

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Substituting values of μ_0 and ϵ_0, $\quad v = \dfrac{1}{\sqrt{4\pi \times 10^{-7} \times 10^{-9}/36\pi}} = 3 \times 10^8$ (m/s)

This is nothing but velocity of light. Thus waves travel in free space with a velocity of light, denoted by c.

$\therefore$

$$v = c = 3 \times 10^8\ \text{(m/s)}.$$

Let us compare the two velocities, velocity in free space and in dielectric. The velocity in dielectric is,

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{1/\sqrt{\mu_0 \epsilon_0}}{\sqrt{\mu_r \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

Almost for all dielectrics, $\mu_r = 1$, giving

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

The number ϵ_r for any dielectric is greater than one, means v is always less than c.

(4) Intrinsic impedance :

$$\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{10^{-9} / 36\pi}} = 120\pi \simeq 377\ (\Omega)$$

(5) Similar to dielectric, waves in free space are TEM waves :

$$\bar{E} \cdot \bar{H} = 0 \quad \text{and}$$

$$\bar{E} \times \bar{H} \Rightarrow \text{in the direction of propagation.}$$

(6) Relation between field components travelling in +x direction :

$$\frac{E_y}{H_z} = -\frac{E_z}{H_y} = 120\,\pi$$

(7) Hemholtz equations :

$$\nabla^2 \bar{E} = -\omega^2 \mu_0 \epsilon_0 \bar{E} \text{ and } \nabla^2 \bar{H} = -\omega^2 \mu_0 \epsilon_0 \bar{H}$$

(8) The properties of wave :

$$\text{Phase constant, } \beta = \omega \sqrt{\mu_0 \epsilon_0}$$

$$\text{Phase velocity, } v = c = \frac{\omega}{\beta}$$

$$\text{Wavelength, } \lambda = \frac{2\lambda}{\beta} = \frac{c}{f}$$

(9) Attenuation :

Free space also have $\sigma = 0$ giving attenuation $\alpha = 0$.

9.3.1 Solved Examples on Plane Wave Propagation in Lossless Media :

Important Formulae

$\beta = \omega \sqrt{\mu \epsilon}$, $\alpha = 0$ $\qquad$ $v = \omega / \beta = 1/\sqrt{\mu \epsilon}$

$\lambda = 2\pi/\beta = v/f$ $\qquad$ $\eta = \sqrt{\mu \epsilon}$

Example 9.3.1 : If $E_y = E_0\, e^{m(x-Ct)}$ show that it is a wave equation, if m is a real number.

Solution : Given expression for E_y represents a wave in $+x$ direction, for which

$$E_x = H_x = 0 \quad \text{and} \quad \partial/\partial y = \partial/\partial z = 0$$

The general wave equation is

$$\nabla^2 \bar{E} = \mu \in \ddot{\bar{E}}$$

The term 'C' in the expression represents the velocity in free space. That is, wave is in free space, for which $\mu = \mu_0$ and $\in = \in_0$. Then the above equation now reduces to

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \in_0 \frac{\partial^2 E_y}{\partial t^2}$$

This is a wave equation for given direction of propagation. If the given E_y represents a wave then it has to satisfy above equation.

To obtain LHS :

$$\frac{\partial E_y}{\partial x} = m\, E_0\, e^{m(x-Ct)}$$

$$\therefore \quad \text{LHS} = \frac{\partial^2 E_y}{\partial x^2} = m^2 E_0\, e^{m(x-Ct)} = m^2\, E_y \quad \text{...Ans.}$$

To obtain RHS :

$$\frac{\partial E_y}{\partial t} = -m\, C\, E_0\, e^{m(x-Ct)}$$

$$\frac{\partial^2 E_y}{\partial t^2} = +m^2 C^2\, E_0\, e^{m(x-Ct)}$$

$$\therefore \quad \text{RHS} = \mu_0 \in_0 \frac{\partial^2 E_y}{\partial t^2} = \mu_0 \in_0 m^2 C^2\, E_0\, e^{m(x-Ct)} = \mu_0 \in_0 m^2 C^2 E_y$$

We know

$$C = \frac{1}{\sqrt{\mu_0 \in_0}}$$

$$\text{RHS} = \frac{1}{C^2}\, m^2 C^2\, E_y = m^2\, E_0\, e^{m(x-Ct)} \quad \text{...Ans.}$$

Thus LHS = RHS. Hence given E_y represents a wave equation.

Example 9.3.2 : Show that function $F = e^{-\alpha z} \sin \frac{\omega}{v}(x - vt)$ satisfies the wave equation $\nabla^2 F = (1/C^2)\, \ddot{F}$ provided that the wave velocity is given by :

$$v = C\left(1 + \frac{a^2 C^2}{\omega^2}\right)^{-1/2}$$

Solution : As F is a function of x and z,

$$\nabla^2 F = \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial z^2}$$

Now $$\frac{\partial^2 F}{\partial x^2} = -\left(\frac{\omega}{v}\right)\left(\frac{\omega}{v}\right) e^{-\alpha z} \sin\frac{\omega}{v}(x - vt) = -\frac{\omega^2}{v^2} F$$

and $$\frac{\partial^2 F}{\partial z^2} = (-\alpha)(-\alpha) e^{-\alpha z} \sin\frac{\omega}{v}(x - vt) = \alpha^2 F$$

$\therefore$ $$\nabla^2 F = \left(\alpha^2 - \frac{\omega^2}{v^2}\right) F$$

Also $\dot{F} = -\omega e^{-\alpha z} \cos\frac{\omega}{v}(x - vt)$ and $\ddot{F} = -\omega^2 e^{-\alpha z} \sin\frac{\omega}{v}(x - vt) = -\omega^2 F$

The given wave equation $$\nabla^2 F = \frac{1}{C^2}\ddot{F}$$

Then $$\left(\alpha^2 - \frac{\omega^2}{v^2}\right) F = \frac{1}{C^2}(-\omega^2) F$$

i.e. $$\alpha^2 - \frac{\omega^2}{v^2} + \frac{\omega^2}{C^2} = 0$$

or $$v^2 = \frac{\omega^2 C^2}{\alpha^2 C^2 + \omega^2} = \frac{C^2}{(\alpha^2 C^2/\omega^2) + 1}$$

or $$v = \frac{C}{\sqrt{1 + (\alpha^2 C^2/\omega^2)}} = C\left(1 + \frac{\alpha^2 C^2}{\omega^2}\right)^{-1/2}$$ ...Ans.

Example 9.3.3 : In free space, $E(z, t) = 10^3 \sin(\omega t - \beta z)\, \bar{a}_y$ (V / m). Obtain $H(z, t)$.

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Solution : $(\omega t - \beta z)$ indicates that wave is travelling in +z direction (negative sign in $-\beta z$ refers to positive direction of propagation while z refers to the direction of propagation).

For +z direction we have, $$\frac{E_x}{H_y} = -\frac{E_y}{H_x} = \eta$$

i.e. $$E_x = \eta H_y \quad \text{and} \quad E_y = -\eta H_x$$

Since given electric field has E_y component, it will result in H_x.

$\therefore$ $$\frac{E_y}{H_x} = -\eta$$

or $$H_x = -\frac{E_y}{\eta} = -\frac{E_y}{120\pi} \quad \text{...}(\eta = 120\pi, \text{ for free space})$$

$$= \frac{-10^{-3}}{120\pi} \sin(\omega t - \beta z) = -2.65 \sin(\omega t - \beta z) \text{ (A/m)}$$ ...Ans.

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Example 9.3.4 : An electric field $\bar{E}$ in free space is given as,

$$\bar{E} = 1000 \cos(10^8 t - \beta y)\, \bar{a}_z \text{ (V/m)}$$

Find β, λ, $\bar{H}$ at P (0.2, 2.5, 0.4) at 10 nsec.

Solution :

Given : $\bar{E} = 1000 \cos(10^8 t - \beta y)\, \bar{a}_z = E_z(y, t)\, \bar{a}_z$

Where $E_z = 1000\,(10^8 t - \beta y)$

From which we get, $\omega = 10^8\,(\text{rad/s})$

Also in free space, $v = c = 3 \times 10^8\,(\text{m/s})$ and $\eta = 120\,\pi\,(\Omega)$

(i) **To find β :**

$$\beta = \frac{\omega}{v} = \frac{10^8}{3 \times 10^8} = 1/3\,(\text{rad/m})$$ ...Ans.

(ii) **To find λ :**

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{(1/3)} = 6\pi = 18.85\,(\text{m})$$ ...Ans.

(iii) **To find $\bar{H}$:**

Given expression for E has only E_z component, travelling in + y direction. The electric and magnetic field components for wave in y direction are related by,

$$\frac{E_z}{H_x} = -\frac{E_x}{H_z} = \eta$$

Since E has only E_z component, it gives only H_x.

$$\therefore \quad H_x = \frac{E_z}{\eta} = \frac{1000\cos(10^8 t - \beta y)}{120\,\pi} = 2.65\cos(10^8 t - \beta y)$$

$$\therefore \quad \bar{H} = H_x\,\bar{a}_x = 2.65\cos(10^8 t - \beta y)\,\bar{a}_x$$

$\bar{H}$ at P(0.2, 2.5, 0.4) and t = 10 ns is

$$\bar{H} = 2.65\cos\left(10^8 \times 10 \times 10^{-9} - \frac{1}{3} \times (2.5)\right)\bar{a}_x$$

$$= 2.65\cos(1 - 0.833)\,\bar{a}_x = 2.65\cos(0.167)\,\bar{a}_x$$

The number 0.167 in the bracket is in radians. Converting in degrees

$$0.167\text{ rad} = 0.167 \times \frac{180}{\pi} = 9.568^\circ$$

$$\therefore \quad \bar{H} = 2.65\cos(9.568^\circ)\,\bar{a}_x = 2.61\,\bar{a}_x\,(\text{A/m})$$ ...Ans.

Example 9.3.5 : An $\bar{E}$ field in free space is

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$$\bar{E} = 800\cos(10^8 t - \beta y)\,\bar{a}_z \quad (\text{V/m})$$

Find i) β, ii) λ, iii) $\bar{H}$ at P (0.1, 1.5, 0.4) at t = 8 ns.

Solution :

Given : $\bar{E} = 800\cos(10^8 t - \beta y)\,\bar{a}_z = E_z(y, t)\,\bar{a}_z$

Where $E_z = 800 \cos(10^8 t - \beta y)$

From which we get, $\omega = 10^8$ rad/s

Also in free space, $v = c = 3 \times 10^8$ (m/s), and,

$\eta = 120\,\pi\,(\Omega)$

(i) To find β :

$$\beta = \frac{\omega}{v} = \frac{10^8}{3 \times 10^8} = \frac{1}{3}\text{ (rad/m)}$$

...Ans.

(ii) To find λ :

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{(1/3)} = 6\pi = 18.85\text{ (m)}$$

...Ans.

(iii) To find $\bar{H}$:

From given $\bar{E}$ it is clear that wave is propagating in +y direction and E has E_z component. For a wave in y direction, E and H are related by

$$\frac{E_z}{H_x} = -\frac{E_x}{H_z} = \eta$$

Since E has only E_z component, it gives only H_x.

$$\frac{E_z}{H_x} = \eta \text{ or } H_x = \frac{E_z}{\eta} = \frac{800 \cos(10^8 t - \beta y)}{120\pi}$$

$\therefore$ $$H_x = 2.122 \cos(10^8 t - \beta y)\text{ (A/m)}.$$

$\therefore$ $$\bar{H} = H_x\,\bar{a}_x = 2.122 \cos(10^8 t - \frac{1}{3}y)\,\bar{a}_x$$

$\bar{H}$ at P (0.1, 1.5, 0.4) and t = 8 ns

$$\bar{H} = 2.122 \cos(10^8 \times 8 \times 10^{-9} - \frac{1}{3} \times 1.5)\,\bar{a}_x$$

$$= 2.122 \cos(0.8 - 0.5)\,\bar{a}_x$$

$$= 2.122 \cos(0.3)\,\bar{a}_x$$

Note that 0.3 in the bracket is in radians. Converting in degrees

$$0.3\text{ rad} = 0.3 \times \frac{180}{\pi} = 17.189°$$

$\therefore$ $$\bar{H} = 2.122 \cos(17.189°)\,\bar{a}_x$$

$$= 2.03\,\bar{a}_x\text{ (A/m)}$$

...Ans.

Find: (i) direction of propagation

(ii)

(iii)

(i) To find direction of propagation :

(ii) To find λ, f and ϵ_r :

$$\therefore \qquad f = \frac{10^8}{2\pi} = 0.1592 \times 10^8 = 15.92 \text{ MHz}$$

The wavelength is calculated as, $\lambda = \frac{2\pi}{\beta} = \pi$ (rad).

The expression for β is, $\beta = \omega\sqrt{\mu\epsilon}$

i.e. $2 = 10^8 \sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12} \epsilon_r}$

Squaring both sides $4 = 10^{16} (4\pi \times 10^{-7} \times 8.854 \times 10^{-12} \epsilon_r)$

$\therefore \qquad \epsilon_r = 35.95$ ...Ans.

iii) To find $\bar{H}$:

This is a second method to obtain $\bar{H}$. First method is as done in Ex. 9.3.3.

The intrinsic impendance,

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{4\pi \times 10^{-7}}{35.95 \times 8.854 \times 10^{-12}}} = 62.83 \ (\Omega).$$

The expression for $\bar{H}$ can be written as,

$$\bar{H} = H_0 \sin(10^8 t + 2z) \ \bar{a}_H$$

where, $\eta = \frac{E_0}{H_0}$ ...Ans.

Example 9.3.8 : The electric field in free space is given by

$\bar{E} = 50 \cos(10^8 t + \beta x) \ \bar{a}_y$ (V/m)

(i) Find the direction of propagation

(ii) Calculate β and time it takes to travel a distance of $\lambda/2$.

Solution :

The given expression for E represents a wave travelling in – x direction with $\omega = 10^8$ (rad/s), in free space.

i) To find direction of propagation :

As stated above + βx term represents a wave in – x direction. ...Ans.

ii) To find β and time :

For free space, the velocity $v = c = 3 \times 10^8$ (m/s).

The quantities v, ω and β are related by,

$$v = \frac{\omega}{\beta}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{1}{4} \times \sqrt{\frac{\mu_0}{\epsilon_0}} = 94.28\ (\Omega)$$

This change in η changes H_0 as

$$H_0 = \frac{E_0}{\eta} = \frac{30\pi}{94.28} = 1.0\ (A/m)$$

The velocity of propagation is given by,

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}} \times \frac{1}{\sqrt{\mu_0\epsilon_0}} \times \frac{1}{\sqrt{\epsilon_r}} = 0.75 \times 10^8\ (m/s)$$

Now,

$$\beta = \frac{\omega}{v} = \frac{10^8}{0.75 \times 10^8} = 1.33\ \text{rad/m}$$

...Ans.

Example 9.3.12 : In a homogeneous region where $\mu_r = 1$ and $\epsilon_r = 50$,

$$\bar{E} = 20\,\pi\, e^{j(\omega t - \beta z)}\, \bar{a}_x\ (v/m),$$

$$\bar{B} = \mu_0 H_m\, e^{j(\omega t - \beta z)}\, \bar{a}_y\ (T)$$

Find ω and H_m if the wavelength is 1.78 m.

Test

$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}}$

$\eta = \frac{120\pi}{\sqrt{\epsilon_r}}$

Solution : We have,

$$\frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}} = \frac{120\,\pi}{\sqrt{\epsilon_r}} = \frac{120\,\pi}{\sqrt{50}} = 53.31$$

$$\therefore \quad H = \frac{E}{53.31} = \frac{20\,\pi}{53.31} = 1.18\ (A/m)$$

The velocity of a wave is,

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{50}} = 42.43 \times 10^6\ (m/s)$$

From given wavelength we calculate β as,

$$\beta = \frac{2\pi}{\lambda} = \frac{2\,\pi}{1.78} = 3.53\ (rad/m)$$

The velocity and β are related using,

$$v = \frac{\omega}{\beta}$$

or

$$\omega = v\beta = (42.43 \times 10^6) \times 3.53 = 1.5\ (rad/s.)\ \times 10^8$$

...Ans.

Example 9.3.13 : In a homogeneous nonconducting region where,

$\mu_r = 1$, find ϵ_r and ω if

$$\bar{E} = 30\,\pi\, e^{j(\omega t - (4/3)y)}\, \bar{a}_x \quad \text{and} \quad \bar{H} = 1.0\, e^{j(\omega t - (4/3)y)}\, \bar{a}_x$$

Solution : We have,

$$\frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}} = \frac{120\,\pi}{\sqrt{\epsilon_r}}$$

iv) To find wave components of $\bar{E}$:

Let the electric field be

$$\bar{E} = \bar{E}_1 + \bar{E}_2$$

where,
$$\bar{E}_1 = E_1 \cos(\omega t - z)\, \bar{a}_{E1}$$

$$\bar{E}_2 = E_2 \sin(\omega t - z)\, \bar{a}_{E2}$$

E_1 and E_2 are obtained using,

$$\eta = \frac{E_1}{H_1} \rightarrow E_1 = \eta H_1 = 60\pi(0.1) = 6\pi \text{ (V/m)}$$

and
$$\eta = \frac{E_2}{H_2} \rightarrow E_2 = \eta H_2 = 60\pi(0.5) = 30\pi \text{ (V/m)}$$

The directions $\bar{a}_{E1}$ and $\bar{a}_{E2}$ are obtained using

$$\bar{a}_{E1} \times \bar{a}_{H1} = \bar{a}_\beta$$
$$\bar{a}_{E1} \times (-\bar{a}_x) = \bar{a}_z$$
$$\rightarrow \bar{a}_{E1} = \bar{a}_y$$

$$\bar{a}_{E2} \times \bar{a}_{H2} = \bar{a}_\beta$$
$$\bar{a}_{E2} \times \bar{a}_y = \bar{a}_z$$
$$\rightarrow \bar{a}_{E2} = \bar{a}_x$$

So the electric field intensity is,

$$\bar{E} = \bar{E}_1 + \bar{E}_2$$
$$= 6\pi \cos(\omega t - z)\, \bar{a}_y + 30\pi \sin(\omega t - z)\, \bar{a}_x$$
$$= 6\pi \cos(1.5\times 10^8 t - z)\, \bar{a}_y + 30\pi \sin(1.5 \times 10^8 t - z)\, \bar{a}_x \text{ (V/m)}$$ ...Ans.

Example 9.3.16 : A 150 MHz uniform plane wave in free space is travelling in $\bar{a}_x$ direction. The electric field intensity has a maximum amplitude of

$200\,\bar{a}_y + 400\,\bar{a}_z$ (V/m) at P (10, 30, – 40) at t = 0. Find

i) Phase constant,
ii) Wavelength,
iii) Phase velocity,
iv) Characteristic impedance and
v) E (x, y, z)

Given : $\mu_0 = 4\pi \times 10^{-7}$ and $\epsilon_0 = 10^{-9}/36\pi$

Tut

Solution :

Given : $f = 150 \times 10^6$ Hz

The wave is travelling in free space so velocity is

$$v = c = 3 \times 10^8 \text{ (m/s)}$$

The wave is travelling in x-direction, so

$$\bar{a}_\beta = \bar{a}_x$$

i) **To find phase constant :**

We have,

$$v = \frac{\omega}{\beta} \quad \rightarrow \quad \beta = \frac{\omega}{v} = \frac{2\pi \times 150 \times 10^6}{3 \times 10^8} = \pi \text{ (rad /m)}$$

...Ans.

ii) **To find wavelength :**

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\pi} = 2 \text{ (m)}$$

...Ans.

iii) **To find phase velocity**

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c = 3 \times 10^8 \text{ (m/s)}$$

...Ans.

This value is very obvious since the wave is travelling in free space.

iv) **To find characteristic impedance :**

$$\eta = \frac{\sqrt{\mu}}{\epsilon} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\,\pi\,(\Omega)$$

...Ans.

This is nothing but intrinsic impedance of free space.

v) **To find E :**

Since at t = 0, the amplitude is maximum it means the variation of E is in cosine manner and the wave is in x-direction, so

$$\bar{E} = (200\,\bar{a}_y + 400\,\bar{a}_z)\cos(\omega t - \beta x)$$

Putting the values of ω and β,

$$\bar{E} = (200\,\bar{a}_y + 400\,\bar{a}_z)\cos(300\,\pi \times 10^6\,t - \pi x) \text{ (V/m)}$$

...Ans.

Example 9.3.17 : In a non magnetic medium

$$\bar{E} = 50\cos(10^9 t - 8x)\,\bar{a}_y + 40\sin(10^9 t - 8x)\,\bar{a}_z \text{ (V/m)}$$ find the dielectric constant ϵ_r and the corresponding $\bar{H}$.

lect 2/4/25

Solution :

To find ϵ_r :

From the given field expression

$$\omega = 10^8 \text{ (rad/s)}, \; \beta = 8,$$

and the wave is propagating in + x direction. In a non magnetic medium $\mu_r = 1$. We have

$$\beta = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_0\epsilon_0\epsilon_r} = \frac{\omega}{c}\sqrt{\epsilon_r}$$

i.e.
$$8 = \frac{10^8}{3\times 10^8}\sqrt{\epsilon_r}$$

or
$$\sqrt{\epsilon_r} = 2.4 \rightarrow \epsilon_r = 5.76$$

To find $\bar{H}$:

To find H, we require intrinsic impedance of the media.

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0\epsilon_r}} = \frac{120\,\pi}{\sqrt{\epsilon_r}} = \frac{120\,\pi}{2.4} = 157.1\ (\Omega)$$

The given electric field we split into two components.

$$\bar{E} = \bar{E}_1 + \bar{E}_2$$

where,
$$\bar{E}_1 = 50\cos(10^9\,t - 8x)\,\bar{a}_y = E_1\cos(10^9\,t - 8x)\,\bar{a}_y$$

$$\bar{E}_2 = 40\sin(10^9\,t - 8x)\,\bar{a}_z = E_2\sin(10^9\,t - 8x)\,\bar{a}_z$$

so,
$$\bar{a}_{E1} = \bar{a}_y,\ \bar{a}_{\beta 1} = \bar{a}_x,\ \bar{a}_{E2} = \bar{a}_z,\ \bar{a}_{\beta 2} = \bar{a}_x,\ E_1 = 50,\ E_2 = 40.$$

The components $\bar{E}_1$ and $\bar{E}_2$ will result in $\bar{H}_1$ and $\bar{H}_2$ giving

$$\bar{H} = \bar{H}_1 + \bar{H}_2 = H_1\cos(10^9\,t - 8x)\,\bar{a}_{H1} + H_2\sin(10^9\,t - 8x)\,\bar{a}_{H2}$$

The amplitudes H_1 and H_2 are obtained by,

$$\eta = \frac{E_1}{H_1} = \frac{E_2}{H_2}$$

i.e.
$$H_1 = \frac{E_1}{\eta} = \frac{50}{157.1} = 0.318\ (\text{A/m})$$

$$H_2 = \frac{E_2}{\eta} = \frac{40}{157.1} = 0.255\ (\text{A/m})$$

The directions $\bar{a}_{H1}$ and $\bar{a}_{H2}$ are obtained using

$$\bar{a}_{E1}\times\bar{a}_{H1} = \bar{a}_\beta \qquad \bar{a}_{E2}\times\bar{a}_{H2} = \bar{a}_\beta$$

$$\bar{a}_y\times\bar{a}_{H1} = \bar{a}_x \qquad \bar{a}_z\times\bar{a}_{H2} = \bar{a}_x$$

i.e. $\bar{a}_{H1} = \bar{a}_z$ i.e. $\bar{a}_{H2} = -\bar{a}_y$

Then the total magnetic field is

$$\bar{H} = 0.318\cos(10^9\,t - 8x)\,\bar{a}_z - 0.255\sin(10^9\,t - 8x)\,\bar{a}_y\ (\text{A/m})$$

...Ans.

Example 9.3.18 : A uniform wave in air has

$$\bar{E} = 10 \cos (2\pi \times 10^9 t - \beta z)\, \bar{a}_y$$

Tutorial

i) Calculate β and λ

ii) Sketch the wave at $z = 0, \lambda/4$

Solution :

Given : $\bar{E} = 10 \cos (2\pi \times 10^9 t - \beta z)\, \bar{a}_y$

From this expression,

$$\omega = 2\pi \times 10^9 \text{ (rad/s)} \text{ or } f = 10^9 \text{ (Hz)}.$$

The wave in air has velocity,

$$v = c = 3 \times 10^8 \text{ (m/s)}$$

i) **To calculate β and λ :**

We have, $v = \frac{\omega}{\beta}$

$$\beta = \frac{\omega}{v} = \frac{2\pi \times 10^9}{3 \times 10^8} = 20.94 \text{ (rad/m)}$$

$$\lambda = \frac{2\pi}{\beta} = 0.3 \text{ (m)}$$

...Ans.

ii) **To sketch the wave at $z = 0, \lambda/4$:**

At $z = 0$, $\bar{E} = 10 \cos (2\pi \times 10^9 t)\, \bar{a}_y$

At $z = \lambda/4$, $\bar{E} = 10 \cos \left(2\pi \times 10^9 t - \frac{2\pi}{\lambda} \times \frac{\lambda}{4}\right) \bar{a}_y$

$$= 10 \sin (2\pi \times 10^9 t)\, \bar{a}_y$$

So these plots are simply cosine and sine function plots at $z = 0$ and $\lambda/4$ respectively. The time period of these waveforms.

$$T = \frac{1}{f} = \frac{1}{10^9} = 1 \text{ (ns)}$$

...Ans.

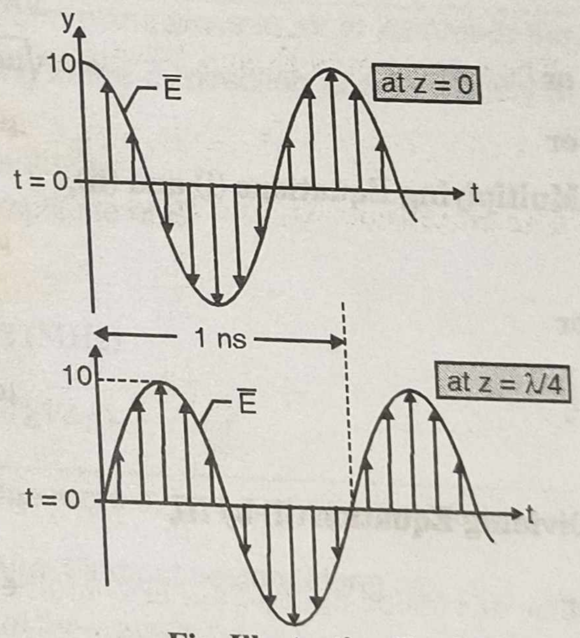

Fig. Illustrating Ex. 9.3.18

Example 9.3.19 : A wave propagating in a lossless dielectric has components

leet

$$\bar{E} = 500 \cos (10^7 t - \beta z)\, \bar{a}_x \text{ (V/m) and}$$

$$\bar{H} = 1.1 \cos (10^7 t - \beta z)\, \bar{a}_y \text{ (A/m).}$$

If the wave is travelling at $v = 0.5c$, find μ_r, ϵ_r, β, λ and η.

Solution :

Given : $\bar{E} = 500 \cos(10^7 t - \beta z)\,\bar{a}_x = E_0 \cos(10^7 t - \beta z)\,\bar{a}_x$

$\bar{H} = 1.1 \cos(10^7 t - \beta z)\,\bar{a}_y = H_0 \cos(10^7 t - \beta z)\,\bar{a}_y$

From these expressions, $\omega = 10^7$ rad/s.

The wave is travelling in + z direction in lossless media.

For lossless media,

$$\eta = \frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}}$$

and

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\epsilon}}$$

Putting the values of E and H,

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{E}{H} = \frac{500}{1.1} = 454.55$$

$$\therefore \quad \frac{\mu}{\epsilon} = 206615.70 \qquad \text{...(i)}$$

Given, velocity $v = 0.5\,c = 0.5 \times 3 \times 10^8 = 1.5 \times 10^8$ m/s

$$\frac{1}{\sqrt{\mu\epsilon}} = 1.5 \times 10^8$$

or

$$\sqrt{\mu\epsilon} = 6.67 \times 10^{-9}$$

or

$$\mu\epsilon = 4.44 \times 10^{-17} \qquad \text{...(ii)}$$

Multiplying Equations (i) and (ii),

$$\mu^2 = 206615.70 \times 4.44 \times 10^{-17} = 9.17 \times 10^{-12}$$

or

$$\mu = 3.028 \times 10^{-6} = \mu_0 \mu_r$$

$$\therefore \quad \mu_r = \frac{3.028 \times 10^{-6}}{4\pi \times 10^{-7}} = 2.390$$

Dividing Equation (ii) by (i),

$$\epsilon^2 = \frac{4.44 \times 10^{-17}}{206615.70} = 2.15 \times 10^{-22}$$

or

$$\epsilon = 1.466 \times 10^{-11} = \epsilon_0 \epsilon_r$$

$$\therefore \quad \epsilon_r = \frac{1.466 \times 10^{-11}}{8.854 \times 10^{-12}} = 1.656$$

To find β we have,

$$v = \frac{\omega}{\beta} \text{ or } \beta = \frac{\omega}{v} = \frac{10^7}{1.5 \times 10^8} = 0.0667 \text{ (rad/m)}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.0667} = 94.25 \text{ (m)}$$

As calculated in the beginning,

$$\eta = 454.55\ (\Omega) \qquad \text{...Ans.}$$